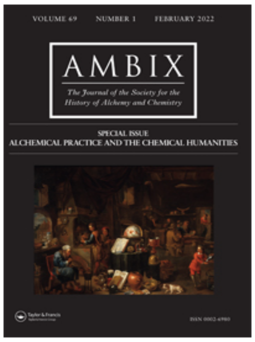

Ambix

ISSN: 0002-6980 (Print) 1745-8234 (Online) Journal homepage: www.tandfonline.com/journals/yamb20

A Machine Reasoning Algorithm for the Digital Analysis of Alchemical Language and its *Decknamen*

Sarah Lang

To cite this article: Sarah Lang (2022) A Machine Reasoning Algorithm for the Digital Analysis of Alchemical Language and its *Decknamen*, Ambix, 69:1, 65-83, DOI: 10.1080/00026980.2022.2038428

To link to this article: https://doi.org/10.1080/00026980.2022.2038428

© 2022 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 11 Mar 2022.

Submit your article to this journal

Article views: 2563

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at https://www.tandfonline.com/action/journalInformation?journalCode=yamb20

AMBIX, Vol. 69 No. 1, February 2022, 65–83

Check for updates

A Machine Reasoning Algorithm for the Digital Analysis of Alchemical Language and its *Decknamen*

SARAH LANG

University of Graz, Austria

Alchemical language has been addressed in several lexicographical studies, historically and recently. This paper discusses the current state of the field and proposes a digital distant-reading approach to the issue of decoding alchemical *Decknamen*. This paper presents an algorithm for the digital analysis of alchemical language using the corpus of printed works by Michael Maier (1568–1622). Alchemical language was used as a medium for negotiating authority, inclusion, and exclusion in alchemical and chymical communities and also as a tool for social and scholarly knowledge-making. This paper addresses the historical problem of understanding early modern alchemical language by computational analysis. Previous studies have applied close reading methodologies to decode *Decknamen*; however, this paper offers a machine reasoning approach to analyse patterns in alchemical language.

Introduction

Contemporary library and special collections efforts have focused heavily on creating digital facsimiles of hand-press texts, made available to researchers as digital images, transcriptions, or even digital editions.[1] This paper asks how textual digitisation efforts can advance the historical study of alchemy beyond merely making texts more easily accessible. How can the complicated language of early modern alchemy benefit from digital analysis? This paper illustrates the opportunities for unpacking the coded and frequently esoteric language of early modern alchemy and chymistry using transcriptions of hand-press alchemical texts. Computational

[1] The field of Digital Humanities refers to digital images of historical objects, i.e. image files, as facsimiles or reproductions, whereas a digital transcription means that the data exist in the form of digital text (like a .txt or other text file format). Machine-processing requires a digital text file, created manually or with Optical Character Recognition (OCR).

© 2022 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution-Non-Commercial-NoDerivatives License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way.

DOI 10.1080/00026980.2022.2038428

methods for text analysis can be tailored to meet the unique needs of historical alchemy research and the specific semantic issues inherent in alchemical coded language in the form of the stylistic device called *Decknamen*. Digital methods can serve as a catalyst for efficient analysis of large alchemical corpora such as that of the German iatrochymist Michael Maier (1568–1622) to address the issues of approaching obscured alchemical language and the particular stylistic devices and cryptographic tropes of early modern alchemy.

The historiography of alchemical language

The alchemical lexicographical endeavour gained traction in the seventeenth century and continues today, from Martin Ruland's famous 1612 alchemical dictionary to Karin Figala and Claus Priesner's late-twentieth-century lexicon.[2] Lexicography is a recurring theme within the history and historiography of alchemy, primarily because alchemical language is a roadblock to comprehending alchemical textual sources for historians today just as much as it was for early modern alchemical practitioners, if not more so.[3] There is an entire genre of publications on alchemical expert language seen as a sub-language (in German *Fachsprache*), particularly in the German research tradition surrounding Paracelsianism.

This tradition, following the footsteps of Gerhard Eis, argues that alchemical language should be understood and treated as an expert sub-language rather than simply specialist terminology. Alchemical language comprises more than technical terms, and also includes phenomena that go beyond the word level, such as stylistic devices or allegories. As such, alchemical *Decknamen* (literally meaning "cover-names") cannot be sufficiently defined by describing them as technical terms.[4] If they were indeed technical terms like those known to us from other disciplines, decoding them would be much more straightforward, and a dictionary of specialist terms would suffice to make their meanings understandable to most readers familiar with the chemical concepts they describe. Yet, as anyone familiar with alchemical texts has experienced, alchemical *Decknamen* are context-sensitive and prone to creative uses and metaphorical neologisms in

[2] Martin Ruland, *Lexicon Alchemiae* (Frankfurt am Main, 1612); Karin Figala and Claus Priesner, "Ein Lexikon der Alchemie – Inhalte und Zielsetzungen," in *Hermetik und Alchemie: Betrachtungen am Ende des 20. Jahrhunderts*, ed. Karin Figala and Helmut Gebelein (Gaggenau: Scientia Nova, 2003), 121–40.

[3] See the pioneering work of Gerhard Eis, "Von der Rede und dem Schweigen der Alchemisten," in *Vor und nach Paracelsus. Untersuchungen über Hohenheims Traditionsverbundenheit und Nachrichten über seine Anhänger* (Stuttgart: Gustav Fischer, 1965), 51–73. See also Jörg Barke, *Die Sprache der Chymie. Am Beispiel von vier Drucken aus der Zeit zwischen 1574–1761* (Tübingen: Niemeyer, 1991). On earlier alchemical lexicography see Bruce Moran's review essay of the Figala/Priesner lexicon, Bruce Moran, "Essay Review: Alchemy, Chemistry and the History of Science," *Studies in History and Philosophy of Science* 31 (2000): 711–20.

[4] On *Decknamen* see Lawrence M. Principe, "Robert Boyle's Alchemical Secrecy: Codes, Ciphers and Concealments," *Ambix* 39 (1992): 63–75; William R. Newman, "'*Decknamen* or Pseudochemical Language'? Eirenaeus Philalethes and Carl Jung," *Revue d'histoire des Sciences* 49 (1996): 159–88. For historiographical works see Julius Ruska and E. Wiedemann, "Alchemistische Decknamen," *Beiträge zur Geschichte der Naturwissenschaften LXVII, Sitzungsberichte der Phys.-Med. Sozietät Erlangen* 56 (1924): 17–36; E. J. Holmyard, "Abu' L-Qasim Al-Iraqi," *Isis* 8 (1926): 403–26; Edmund Oskar von Lippmann, *Entstehung und Ausbreitung der Alchemie. Mit einem Anhange: Zur älteren Geschichte der Metalle. Ein Beitrag zur Kulturgeschichte* (Berlin: Springer, 1919), vol. 1.

ways that other technical terms are not. The modern linguistic definition of a technical term is ultimately irreconcilable with how alchemical *Decknamen* function as extremely context dependent cover-words.[5] Thus, lexicography alone cannot fully describe alchemical *Decknamen*.

More recently, James Voelkel has suggested that creating an integrated digital resource could bring the alchemical lexicographical enterprise into the twenty-first century.[6] His proposal represents an important trend in the study of alchemical language as it moves it away from mere lexicography, allowing for a more wholistic approach suited to encoding not only the words used in alchemical texts but also common concepts which can be denoted by different words. Voelkl proposes a Chymical Encyclopedia, Database, and Repository (CEDR): a complex digital system that is a combination encyclopaedia, repository of primary textual sources, and knowledge organisation system (KOS).[7] The proposed CEDR follows the example of the well-known Stanford Encyclopaedia of Philosophy and uses the format of Priesner and Figala's *Lexikon Alchemie,* which is characterised by high-quality expert contributions. Additionally, Voelkel's CEDR would link to the primary source texts in its repository of digital facsimiles, as well as contain a database of historical recipes and video recordings of scholarly replications of chymical experiments.

This paper introduces a digital methodology in response to Voelkel's vision, which integrates the resources of an encyclopaedia and a lexicon with digitised alchemical texts.[8] Combining close reading approaches to alchemical language with resources for the historical context and interpretation of early modern alchemy creates exciting new opportunities. However, for such a system to work, links need to be established between the different media involved. Digital annotation can provide the vital link between specific textual information and resources for alchemical knowledge. Such a system would also allow for the tracking of relationships between terms across a large corpus, ultimately providing scholars with a new approach to analysing alchemical expert knowledge and *Decknamen* in the context of lexicography, while correctly situating the language of alchemy as a sub-language with a unique set of stylistic devices and linguistic practices.

5 It is important to note that from a linguistic perspective semantic ambiguity is a central characteristic of words, as they can have multiple meanings that are context dependant. A modern understanding of a technical term is that it was designed to avoid this semantic ambiguity inherent in natural language which may result in the need to disambiguate from several meanings.

6 James Voelkel et al., *A Proposed Chymical Encyclopedia, Database and Repository* (Philadelphia, PA: Science History Institute, 2018).

7 Since the term "database" used in the original publication refers to only one possible technical implementation of what is being proposed, the term was replaced here by the more fitting umbrella term "Knowledge Organization System" (KOS). A knowledge organisation system is an inclusive term for any digital base of data. Voelkl's promising proposal has not been implemented to this day.

8 These two terms are often conflated, however they differ in that an "encyclopaedia" summarises comprehensive knowledge about a topic, whereas a "lexicon" is a reference for linguistic information.

The corpus of Michael Maier can be used as an example of what such a unified resource could look like and the benefits it would have for scholarship. While there are a wide range of methods within digital humanities to approach such an undertaking, the most appropriate for the issue of alchemical terminology would be a digital ontology to describe class relationships rather than linguistic properties of terms found in a traditional lexicon.[9] A digital ontology would allow for storing information that goes beyond linguistic description, and it makes it possible to trace a concept across languages. It can also generate a grid into which future digital information can be stored as Linked Open Data (LOD), such as a database of recipes and historical-critical recreations of alchemical experiments, as has been proposed by Voelkel. The goal of LOD within digital humanities scholarship is to reconcile different data sources on related topics. It can even allow for indirect relations between tokens of information to be made explicit; for example: Concept A from data provider B is similar but not identical to concept C from data provider D.

Ultimately, this paper argues that rather than create yet another dictionary or encyclopaedia of alchemical terms, as plenty of such resources have been created since the seventeenth century, scholarly efforts should be put toward developing a resource to connect such information and integrate the textual context regarding its use. Such a knowledge organisation system would bring together existing resources that when referenced individually, only supply a portion of the complex and layered symbolism in alchemical language.[10] Although language is a central aspect in creating alchemical knowledge, there has not been a comprehensive study on alchemical language beyond a few theoretical essays.[11] As a solution,

[9] In the context of information science, a digital ontology is defined as "an explicit specification of a conceptualisation. The term is borrowed from philosophy, where an ontology is a systematic account of Existence. For knowledge-based systems, what 'exists' is exactly that which can be represented." (Tom Gruber, "A Translation Approach to Portable Ontology Specifications," *Knowledge Acquisition* 5 (1993): 199–220, on 199.) In the case of alchemy research, a digital ontology is a knowledge model that focuses on this domain of interest. Examples for technological implementations of such ontologies are standards like RDF (Resource Description Framework), RDFS (RDF-Schema), or SKOS (Simple Knowledge Organisation System) that use different representation forms and contain statements, mostly in the form of describing class relationships (such as "a chihuahua is a dog," "a dog is an animal") which can be expressed in different data structures, such as Turtle syntax (https://www.w3.org/TR/turtle/) or XML (eXtensible Markup Language).

[10] For more on the KOS proposed here see Sarah Lang, "Vom 'Wissen in Buchform' zum formalen Wissensmodell. Digitale Aufbereitung wissenschaftshistorischer Drucke am Beispiel der Alchemica Michael Maiers," in *Wissen und Buchgestalt*, ed. Philipp Hegel and Michael Krewet (Wiesbaden: Harrassowitz Verlag, forthcoming). See also Lang, "Digital Scholarly Editions of Alchemical Texts as Tools for Interpretation," in *Digitale Edition in Österreich*, ed. Helmut Klug and Roman Bleier (Graz, forthcoming); "Digitale Erschließungsmethoden für alchemische Texte am Beispiel der Symbola Aureae Mensae Michael Maiers," *Alchemie – Genealogie und Terminologie, Bilder, Techniken und Artefakte*, ed. Petra Feuerstein-Herz and Ute Frietsch (Wiesbaden: Harrassowitz, 2021); "Digitale Annotation alchemischer Decknamen. Die Allegoriae werden uns nit mehr verborgen seyn," in *Annotations in Scholarly Editions and Research: Functions, Differentiation, Systematization*, ed. Julia Nantke and Frederik Schlupkothen (Berlin: De Gruyter, 2020), 201–19, https://doi.org/10.1515/9783110689112-010 (accessed 1 January 2022).

[11] For earlier works on alchemical language see J. R. Partington, "Report of Discussion Upon Chemical and Alchemical Symbolism," *Ambix* 1 (1937): 61–4; Michel Butor, *Die Alchemie und ihre Sprache*. (Frankfurt: Fischer, 1990); Ruska and Wiedemann, "Alchemistische Decknamen," 17–36; Alfred Siggel, *Decknamen in der arabischen alchemistischen Literatur. Deutsche Akademie der Wissenschaften zu Berlin. Institut für Orientforschung 5* (Berlin: Akademie-Verlag, 1951); von Lippmann, *Entstehung und Ausbreitung der Alchemie*; Michel Foucault, "La prose du monde," in *Les Mots et Les Choses. Une Archéologie des Sciences Humaines* (Paris: Gallimard, 1966), 32–57; Umberto Eco, "Il Discorso Alchemico E Il Segreto Differito," in *I Limiti Dell'interpretazione* (Milano: La nave di Teseo, 2016), 97–116; Aleida Assmann, *Im Dickicht der Zeichen* (Berlin: Suhrkamp, 2015).

this paper proposes a digital approach to model alchemical language through digital annotation.

Shifting views on alchemical language

When attempting to address *the* language of *the* alchemical tradition caution must be exercised, as there are multiple distinct, albeit not mutually independent, alchemical traditions that should not be treated as a homogeneous whole.[12] Yet, if there is one commonality between the various alchemical traditions, it would be the characteristic style of communication that can be described as "alchemical language." Expert languages, especially the language of alchemists and chymists, can be understood as means of access control by limiting the audience and excluding those who do not understand the particular mode of functioning of the linguistic code thus creating a sense of community among those in the know.[13] The (frequently playful) rhetoric of secrecy and the use of unnecessarily complicated stylistic and linguistic devices can be seen as a means of self-fashioning as an alchemical expert.

There are examples of extensive alchemical riddling to the point that the contents of the encoded text hardly further the alchemical argument and are sometimes entirely irrelevant to the theoretical background of the author in question.[14] Beyond the performative aspect of obfuscating language, the means of communication also shape its meaning. The use of certain allegories to approach chemical reactions can be understood as a very practical visualisation tool for communicating details of chemical processes at a time when exact measurement was not available. However, overuse of alchemical *Decknamen* can lead to misunderstandings when, for instance, an analogy is not a perfect fit for the subject matter. Conversely, deliberately obscuring language and the resulting need for interpretation can drive creativity and innovation in research and experimentation, as exemplified by the *Processus Universalis* recipe group based on an unclear process described by Polish alchemist Michael Sendivogius (1566–1636).[15]

The historiography of alchemical language has contributed to the confusion surrounding *Decknamen* today. Umberto Eco and Hans-Werner Schütt have presented "obscurantist" theories of alchemical language, arguing that alchemical language is ultimately a riddle for which no systematic solution can be formulated, or to which

12 See Lawrence M. Principe and William R. Newman, "Some Problems with the Historiography of Alchemy," in *Secrets of Nature: Astrology and Alchemy in Early Modern Europe*, ed. William R. Newman and Anthony Grafton (Cambridge, MA: MIT Press, 2001), 385–432.

13 Koen Vermeir, "Openness Versus Secrecy? Historical and Historiographical Remarks," *The British Journal for the History of Science*, Special Issue 45 (2012): 165–88, on 179.

14 William R. Newman, "'*Decknamen* or Pseudochemical Language'?"

15 Thomas Moenius, Alexander Kraft and Gerhard Görmar, "Andreas Orthelius und der Processus Universalis"; Rainer Werthmann and Christian-Heinrich Wunderlich, "Eine Rekonstruktion alchemischer Laborprozesse am Beispiel der Processus Universalis Rezeptgruppe" in Alchemische Labore. Texte, Praktiken und materielle Hinterlassenschaften / Alchemical Laboratories. Texts, practices, material relics, ed. Sarah Lang et al. (Graz: Grazer Universitätsverlag, 2022).

there is no solution at all.[16] Schütt speaks of "layers" of alchemical language, whereas Eco has popularised the concept of "hermetic semiosis," i.e. the idea that alchemical language is profoundly void of meaning and will only cause the reader to err indefinitely in a spiral of semiosis, postulating the existence a specific type of "alchemical discourse." While Schütt references Eco's previous work, their theories are built from different sources. Both identify alleged characteristics of alchemical language by citing quotations from alchemical texts entirely different in character, conventions and genre, of one to two sentences in length and ranging across an extremely long time-span, from the fourth-century writings of Zosimos to the twentieth-century Italian philosopher Evola. One of the main reasons this approach is problematic is that it conflates quite distinct parts of the alchemical tradition with disparate cultural practices and constraints, despite their highly varied historical contexts. Additionally, some of the authors referenced by Schütt and Eco are considered today to be part of the reception of the history of alchemy, rather than personally contributing to the history of alchemy and chymistry (such as Evola). Accordingly, none of the resulting theoretical frameworks accurately represent any individual piece of alchemical writing, rendering the results quite useless for practical application. Due to the considerable influence and credibility of Eco as a prominent semiotician, opinions inspired by his theory of "hermetic semiosis" still haunt the popular historiography of alchemical language to this day, not unlike Jungian interpretations of alchemy, despite the fact that those particular theories cannot be reconciled with contemporary standards of historical methodology.

Historiographic issues surrounding alchemical language have been addressed and remedied in Lawrence Principe and William Newman's "New Historiography of Alchemy."[17] While not explicitly focusing on alchemical language, Principe and Newman engage in discussions of *Decknamen* and identify different encryption techniques to present a methodology based on chemical experimentation to decode secretive linguistic devices. Not only have they called upon scholars to stop using anachronistic terminology, inviting them to use terms like "chymistry" to discuss more accurately early modern (al)chemical practice, they have also made a convincing case against strictly spiritual interpretations of alchemy, ultimately creating a means for an unbiased, thorough analysis of alchemical language, which insists on descriptive precision.[18]

[16] Umberto Eco, "Il Discorso Alchemico E Il Segreto Differito"; Hans-Werner Schütt, "Sprachschichten der Alchemie," *Berichte zur Wissenschaftsgeschichte* 17 (1994): 89–99.

[17] On the "New Historiography of Alchemy" see Principe and Newman, "Some Problems," 385–432; Marcos Martinón-Torres, "Some Recent Developments in the Historiography of Alchemy," *Ambix* 58 (2011): 215–37.

[18] For example, see Principe and Newman, "Some Problems," 385–432; Newman, "'*Decknamen* or Pseudochemical Language'?" For earlier works touching on alchemical language, see: J.R. Partington, "Report of Discussion Upon Chemical and Alchemical Symbolism," Ambix 1 (1937): 61–64; Michel Butor, *Die Alchemie und ihre Sprache. Original in: "Répertoire I", Les Editions de Minuit, Paris 1960* (Frankfurt: Fischer 1990); Julius Ruska and E. Wiedemann, "Alchemistische Decknamen," *Beiträge zur Geschichte der Naturwissenschaften LXVII, Sitzungsberichte der Phys.-Med. Sozietät Erlangen* 56 (1924): 17–36; Alfred Siggel, *Decknamen in der arabischen alchemistischen Literatur.* Deutsche Akademie der Wissenschaften zu Berlin. Institut für Orientforschung 5 (Berlin: Akademie-Verlag,

Two other major developments in the history of science and alchemy challenge long-held assumptions about alchemical language concerning secrecy and the Scientific Revolution. Scholarship on the transformation of chymical language from secrecy to openness during the paradigm shifts of the seventeenth century has been personified by figures such as Andreas Libavius (1550–1616) and Robert Boyle (1627–1691).[19] However, the validity of a dichotomy between secrecy and openness has been called into question by scholarship from both history of science as well as secrecy studies. The early modern rhetoric of secrecy and the "economy of secrets" has been re-evaluated in recent scholarship.[20] Additionally, a singular Scientific Revolution is now widely contested by historians of science. Discussions of the historiographic issue of alchemical language and how it differs from the language of chemistry have, until relatively recently, relied heavily on a supposed dichotomy of secrecy versus openness and the concept of a disruption caused by the Scientific Revolution. Recent scholarship has traded the teleological narrative of a rejection of the secretive alchemical language in favour of a new chemical language for one that centres historical actors engaging in the practice of secretive alchemical language.[21]

The fact that Principe, Newman, and others have been able to successfully decode *Decknamen*, and, in turn, evaluate the experimental side of the practical recipes hidden within them, validates the philological decoding of *Decknamen* through experimentation in a laboratory, and illustrates why outdated "obscurantist" theories of alchemical language can no longer stand. Newman and Principe's success in developing a straightforward methodology that can be applied to texts and re-creations of experiments beyond the examples from which it was developed, proves that alchemical language was created with the intention of comprehensibility, at least for adepts. In the light of these developments, long-held assumptions about alchemical language based on now-outdated theories must be called into question, and a new theory of alchemical language needs to be developed that builds on the methodologies that emerged from the "New Historiography of Alchemy."

Approaching the issue of alchemical language

Scholars have employed several methodologies to address the problem of interpreting alchemical language, which can be categorised as

[18] *Continued*
1951); Edmund Oskar von Lippmann, *Entstehung und Ausbreitung der Alchemie. Mit einem Anhange: Zur älteren Geschichte der Metalle. Ein Beitrag zur Kulturgeschichte. vol. I* (Berlin: Springer, 1919); Chapter "La prose du monde", Michel Foucault, *Les Mots et Les Choses. Une Archéologie des Sciences Humaines* (Paris: Gallimard, 1966), 32–57; Umberto Eco, "Il Discorso Alchemico E Il Segreto Differito".; Aleida Assmann, Im Dickicht der Zeichen (Berlin, 2015).

[19] See Owen Hannaway, *Chemists and the Word: The Didactic Origins of Chemistry* (Baltimore, MD: Johns Hopkins University Press, 1975); Jan Golinski, "Chemistry in the Scientific Revolution: Problems of Language and Communication," in *Reappraisals of the Scientific Revolution*, ed. David C. Lindberg and Robert S. Westman (Cambridge: Cambridge University Press, 1990), 367–96.

[20] For example see Koen Vermeir and Dániel Margócsy, "States of Secrecy: An Introduction," *The British Journal for the History of Science*, Special Issue 45 (2012): 153–64.

[21] See Golinski, "Chemistry in the Scientific Revolution."

(1) *Theorise*: Devise a theory as done by Eco or Schütt.[22] However, as seen in past attempts to theorise alchemical language, methodological fallacies must be avoided and the acute danger of creating a theory which is not applicable to actual historical sources.
(2) *Close Reading*: Analyse textual examples from a single author. This approach provides more concrete evidence than a generalised theory; however, it is difficult to draw conclusions about the entire phenomenon of "alchemical language" from a single source or author.
(3) *Dictionaries*: Examine the effects of confusion caused by alchemical language through the curation of information. The existence of such dictionaries alludes to the urge to organise an overwhelming amount of complexity within alchemical language and practice. Alchemical dictionaries have their own history and can be organised into three source types:

    (a) Seventeenth-century dictionaries of alchemy[23]
    (b) Modern dictionaries appeared in the early twentieth century, and a revival of this type in the 1990s[24]
    (c) Online resources since the 2000s, not all of which are dictionaries in the strictest sense of the term[25]

    A limitation to the dictionary approach is that alchemical *Decknamen* require that the textual semantic context be properly understood. Individual words do not necessarily determine meaning, but rather their meaning is derived from how they are grouped.

(4) *Meta-analysis*: Look at the beliefs of historical actors practicing alchemy by contextualising historical accounts explaining the functioning of alchemical language.

[22] Eco, "Il Discorso Alchemico E Il Segreto Differito"; Hans-Werner Schütt, "Sprachschichten der Alchemie."

[23] See Wilhelm Johnson, *Lexicon Chimicum* (London, 1657); Ruland, *Lexicon*; William R. Newman, "Index Chemicus Ordinatus," in *The Chymistry of Isaac Newton*, ed. Newman (Bloomington: Indiana University, 2009), https://webapp1.dlib.indiana.edu/newton/indexChemicus.do (accessed 3 February 2021); Antoine-Joseph Pernety, *Dictionnaire Mytho-Hermétique* (Paris, 1758).

[24] The following list is by no means comprehensive: Fritz Lüdy, *Alchemistische und chemische Zeichen* (Stuttgart: Gesellschaft für Geschichte der Pharmazie, 1928); J. R. Partington, "Report of Discussion Upon Chemical and Alchemical Symbolism," *Ambix* 1 (1937): 61–4; Wolfgang Schneider, *Lexikon alchemistisch-pharmazeutischer Symbole* (Weinheim: Verlag Chemie, 1962); Gustav Wilhelm Geßmann, *Die Geheimsymbole der Alchymie, Arzneikunde, Astrologie*, reprint (Ulm: Verlag Haug, 1960) and *Die Geheimsymbole der Chemie und Medicin des Mittelalters. Eine Zusammenstellung der von den Mystikern und Alchymisten gebrauchten geheimen Zeichenschrift, nebst einem kurzgefassten geheimwissenschaftlichen Lexikon*, reprint (Graz: Dr. Martin Sändig Gmbh, 1979); Fred Gettings, *Dictionary of Occult, Hermetic and Alchemical Sigils* (London: Routledge, 1981); Mark Haeffner, *A Dictionary of Alchemy: From Maria Prophetissa to Isaac Newton* (London: The Aquarian Press/Harper Collins, 1991); Lyndy Abraham, *A Dictionary of Alchemical Imagery* (Cambridge: Cambridge University Press, 1998); Diana Fernando, *The Dictionary of Alchemy. An A–Z of History, People and Definitions* (London: Vega, 2002); Jordan Stratford, *A Dictionary of Western Alchemy* (Wheaton, IL: Quest Books Theosophical Publishing House, 2011).

[25] William R. Newman, "Alchemical Glossary," in *The Chymistry of Isaac Newton,* ed. Newman (Bloomington: Indiana University, 2009), https://webapp1.dlib.indiana.edu/newton/reference/glossary.do (accessed 3 February 2021) and "Index Chemicus Ordinatus," https://webapp1.dlib.indiana.edu/newton/indexChemicus.do (accessed 3 February 2021); Figala and Priesner, "Ein Lexikon der Alchemie"; Ute Frietsch, "Alchemie Thesaurus," *Herzog August Bibliothek Wolfenbüttel*, 2017, http://alchemie.hab.de/thesaurus (accessed 3 February 2021) and Ute Frietsch, "Fachgruppen," http://alchemie.hab.de/alchemische-fachgruppen (accessed 3 February 2021).

(5) *Cryptanalysis*: Examine the obscured communication typical of alchemical language and learn how it functions cryptologically. This can be achieved by analysing phenomena such as *Decknamen*.

A digital approach to the corpus of Michael Maier as a case study

In today's information age, the topic of social knowledge has become a relevant issue for sociological and philosophical epistemology studies.[26] Within the history of science, interest in information science is predominately regarding early modern anxieties of "information overload."[27] In the historiography of alchemy, discussions of tacit and social knowledge are usually within the context of craft or popular knowledge.[28] Regarding the sociology of tacit knowledge in the specific context of scientific expertise, Harry Collins has provided a thorough update to Michael Polanyi's mid-twentieth-century theory of tacit knowledge and thereby offers a sound theoretical base for its application to information and computer science, specifically addressing the problems of how tacit knowledge can or cannot be made explicit in computer strings and data.[29]

While alchemical symbols could theoretically be arbitrary,[30] the order in which they are grouped is intentional, and this is how the symbols derive meaning.[31] This same linguistic principle has been identified by scholars with regard to the problem of word-sense disambiguation in general – a word can be recognised by the company it keeps, and thus the meaning of a word can be determined by

[26] On "social knowledge" see Everett Mendelsohn, "The Social Construction of Scientific Knowledge," in *The Social Production of Scientific Knowledge*, ed. Everett Mendelsohn, Peter Weingart, and Richard Whitley (Dordrecht: Springer, 1977), 3–26; Helen E. Longino, *Science as Social Knowledge: Values and Objectivity in Scientific Inquiry* (Princeton, NJ: Princeton University Press, 1990); Alvin I. Goldman, "Chapter 8: Science," in *Knowledge in a Social World* (Oxford: Oxford University Press, 1999), 221–71.

[27] See Ann M. Blair, "Reading Strategies for Coping with Information Overload ca. 1550–1700," *Journal of the History of Ideas*, Special Issue 64 (2003): 11–28; Daniel Rosenberg, "Early Modern Information Overload. Introduction," *Journal of the History of Ideas*, Special Issue 64 (2003): 1–9; Brian W. Ogilvie, "The Many Books of Nature: Renaissance Naturalists and Information Overload," *Journal of the History of Ideas*, Special Issue 64 (2003): 29–40; Arndt Brendecke, "Papierfluten. Anwachsende Schriftlichkeit als Pluralisierungsfaktor in der frühen Neuzeit," *Mitteilungen des Sonderforschungsbereichs 573 "Pluralisierung und Autorität in der Frühen Neuzeit"* 1 (2006): 21–30; Ann M. Blair, *Too Much to Know: Managing Scholarly Information Before the Modern Age* (New Haven, CT: Yale University Press, 2010); Andreas Mahler and Martin Mulsow, "Einleitung: Die Vielfalt der Ideengeschichte," in *Texte zur Theorie der Ideengeschichte*, ed. Andreas Mahler and Martin Mulsow (Stuttgart: Reclam, 2014), 9–55.

[28] For example see Pamela O. Long, *Openness, Secrecy, Authorship: Technical Arts and the Culture of Knowledge from Antiquity to the Renaissance* (Baltimore, MD: The Johns Hopkins University Press, 2001); Pamela Smith, "What Is a Secret? Secrets and Craft Knowledge in Early Modern Europe," in *Secrets and Knowledge in Medicine and Science, 1500–1800*, ed. Elaine Leong and Alisha Rankin (New York, NY: Routledge, 2016), 47–68.

[29] Michael Polanyi, *The Tacit Dimension (1966): With a New Foreword by Amartya Sen* (Chicago, IL: Chicago University Press, 2009); Harry Collins, *Tacit and Explicit Knowledge* (Chicago, IL: Chicago University Press, 2010).

[30] Alchemical symbols are usually arranged around analogies that do have a system, common theme, or motif unlike cryptological ciphers or codes, which involve the arbitrary switching out of letters or words respectively. See Barbara Obrist, "Alchimie et Allégorie Scripturaire Au Moyen Age," in *Allégorie des Poètes, Allégorie des Philosophes. Études sur la poétique et l'herméneutique de l'allégorie de l'Antiquité à la Réforme*, ed. Gilbert Dahan and Richard Goulet (Paris: Librairie Philosophique J. Vrin, 2005), 245–65.

[31] Butor, *Die Alchemie und ihre Sprache*, 22.

looking at its direct linguistic context.[32] In computer linguistics the most common method of such concordancing is referred to as a "keyword in context" (KWIC). This paper proposes that alchemical *Decknamen* can be disambiguated using a variation of this principle. However, instead of investigating the linguistic context of the words directly surrounding a term, a computer approach to alchemical *Decknamen* would examine the "context of *Decknamen*" via a KOS containing all instances of alchemical expert vocabulary surrounding a word. Such a "context of *Decknamen*" could mean digitally annotating the five alchemical terms that precede and follow the term in question and subsequently encoding them in a digital KOS to allow for a computer to perform simple machine reasoning tasks. This would require a certain amount of historical context, such as "Michael Maier was an iatrochemist" or "Hermes Trismegistos is a mythological figure" expressed in the form of subject-verb-object relations (X is a Y, for example) to encode in a Resource Description Framework[33] (RDF)-based digital ontology. The more difficult an example is to disambiguate, the more relations would need to be written in the form of the aforementioned statements for the machine to have adequate information to proceed.

What follows is a proposal for a new digital approach building on the methodology and arguments put forth by Voelkel, Newman, and Principe. Alchemical language and its groupings are a social consensus made by a collection of experts. The key to the understanding of alchemical *Decknamen* is shared (practical) chemical knowledge and experience.[34] The methodology put forth in the aforementioned studies by Principe and Newman requires experimental re-enactment. In the many instances in which recreating an alchemical experiment is not possible, a similar proof of coherence may be achieved via digital methods using logic and machine reasoning. Given the limited availability of historical chemical expertise and resources for experiment, scholarship would benefit from a less resource-intensive alternative that does not require expert chemical knowledge at all stages of the interpretation process.

This approach uses a digital KOS encoded as a digital semantic web ontology that models the grouping of alchemical vocabulary and *Decknamen* using digital annotation, allowing for creating a descriptive model. Alchemical vocabulary, including its use and groupings, can be traced in a digital corpus using digital polysemantic annotation while simultaneously conserving the integrity of the original text.[35] A descriptive modelling approach using digital annotation can make textual phenomena traceable and, thus, more convenient to analyse. Once a large enough dataset is

32 Willard McCarty, "Beyond the Word: Modelling Literary Context," in *Text Technology*, ed. Lisa Charlong, Alan Burke, and Brad Nickerson (2007), http://www.mccarty.org.uk/essays/McCarty,%20Beyond%20the%20word.pdf (accessed 1 January 2022).

33 An RDF is a standardised data interchange format for the web: (https://www.w3.org/RDF/).

34 This argument is central to the methodology of the "New Historiography of Alchemy."

35 Many common text mining methods are based on a so-called "bag of words" that disregards the order of the original text. Although scholars have developed methods to circumvent this issue, digital annotation is a more appropriate method for cases in which the integrity of the text is essential.

created that references multiple alchemical and chymical authors, then more accurate conclusions can be drawn about meaning within alchemical language. This paper is not attempting to present new results from specific alchemical authors but proposes a method based on the corpus of Michael Maier as a case study.[36]

There are many reasons why Michael Maier is an appropriate starting point for such an endeavour. The German iatrochymist Michael Maier has attracted considerable attention over the last few years. While the digitised corpus of his printed books yields a total of 3500 PDF pages, his works are "mentioned very often, but read very little."[37] This has resulted in scholarship on Maier relying heavily on secondary literature.[38] Apart from the popular emblem book *Atalanta Fugiens* (1617/8),[39] which was recently published as a digital edition, his oeuvre is relatively under-studied in terms of in-depth textual analysis of his works in the form of critical editions and philological commentaries or *close reading*.[40] However, without a detailed analysis of primary source texts, there is little hope that the state of research criticised by scholars, such as Erik Leibenguth and Volkhard Wels, will get remedied any time soon.[41] Text mining in Maier's corpus could help solve this problem, making Maier a suitable case study for developing a method for digital analysis. Additionally, Maier is encyclopaedic and quite verbose in his writing style, providing a good base for comprehensive digital analysis. Could the proposed *distant reading* shed light on Maier's corpus?[42]

[36] What I propose here is in a similar vein to A. M. Duncan's 1981 article in which he states: "this paper is intended to suggest research ought to be done rather than as a report of completed research." A. M. Duncan, "Styles of Language and Modes of Chemical Thought," *Ambix* 28 (1981): 83–107, on 105.

[37] Volkhard Wels, "Poetischer Hermetismus. Michael Maiers *Atalanta Fugiens* (1617/18)," in *Konzepte des Hermetismus in der Literatur der Frühen Neuzeit*, ed. Peter-André Alt and Volkhard Wels (Göttingen: V & R Unipress, 2010), 149–94, on 149. Wels is especially critical of this, even regarding Maier's most famous and most well researched work, the *Atalanta fugiens*.

[38] Erik Leibenguth stresses this point: "Charakteristisch für die Maierforschung sind mit wenigen […] Ausnahmen eine Abhängigkeit und teils sinnentstellende Rezeption von Sekundärquellen sowie die weitgehende Unkenntnis der lateinischen Primärtexte." Leibenguth, *Hermetische Philosophie des Frühbarock. Die "Cantilenae Intellectuales" Michael Maiers. Edition mit Übersetzung, Kommentar und Bio-Bibliographie* (Tübingen: Niemeyer, 2002), 8–9. Cf. Wels, "Poetischer Hermetismus," 149.

[39] On *Atalanta fugiens* see Tara Nummedal and Donna Bilak (eds.), *Furnace and Fugue. A Digital Edition of Michael Maier's Atalanta Fugiens (1618) with Scholarly Commentary* (Charlottesville: University of Virginia Press, 2020), https://furnaceandfugue.org/ (accessed 3 February 2021).

[40] Lots of research on Maier's vita has been done since the 1990s. On Maier's vita, see: Karin Figala and Ulrich Neumann, "'Author Cui Nomen Hermes Malavici'. New Light on the Biobibliography of Michael Maier (1569–1622)," in *Alchemy and Chemistry in the 16th and 17th Centuries*, ed. Piyo Rattansi and Antonio Clericuzio (Dordrecht: Kluwer Academic Publishers, 1994), 121–48; Erik Leibenguth, *Hermetische Philosophie des Frühbarock*; Hereward Tilton, *The Quest for the Phoenix. Spiritual Alchemy and Rosicrucianism in the Work of Count Michael Maier (1569–1622)* (Berlin/NY: De Gruyter, 2003). Detailed studies exist regarding Maier's doctoral theses (Theses de epilepsia, 1596), Examen (1617), Atalanta Fugiens (1617) und Cantilenae intellectuales (1622). Michael Maier, Atalanta fugiens (Oppenheim 1617); Michael Maier, Examen Fucorum Pseudo-chymicorum Detectorum (Frankfurt 1617); Michael Maier, Cantilenae Intellectuales (Rostock 1622); they are: Wolfgang Beck, *Michael Maiers Examen Fucorum Pseudo-Chymicorum – Eine Schrift wider die falschen Alchemisten*. (München: PhD Thesis, TU München, 1991); Hans Roger Stiehle, *Michael Maierus Holsatus (1569–1622), Alchemist und Arzt. Ein Beitrag zur naturphilosophischen Medizin in seinen Schriften und zu seinem Wissenschaftlichen Qualifikationsprofil* (München: PhD thesis, TU München, 1991); Erik Leibenguth, Hermetische Philosophie des Frühbarock; Furnace and Fugue, ed. Tara Nummedal and Donna Bilak (2020).

[41] Leibenguth, *Hermetische Poesie* and Wels, "Poetischer Hermetismus."

[42] The role of distant reading is not to replace close reading, as Moretti had initially promoted, but to inform a potential reader of pieces of text worth studying in detail, but also keeping enough overview over the corpus as whole to

Modelling Mercurius: a digital semantic web ontology

One reason why a new model must be developed is that existing digital resources simply cannot be applied to such a project because they are not specific enough, such as lexica containing abstract concepts rarely mentioned in actual texts, or are static and thus do not take into account the complex and creative use of alchemical *Decknamen*, which cannot just be "translated" with a dictionary.[43] A second problem arises in the process of digital modelling and KOS – that the historical terms are composed of strings of letters, whereas a digital ontology deals with concepts. It is challenging to map concrete linguistic strings onto more abstract concepts contained within digital KOS.[44] To circumvent this issue, entries for the digital ontology are generated using the detailed indices of Maier himself, which is supplemented by an algorithm to detect auto-semantics, or content words, likely to be relevant to alchemical vocabulary. Then, a concordance is created for the occurrences of those terms. Following the logic of the "Keyword in Context," a word derives meaning from the company it keeps. This principle can be leveraged to disambiguate alchemical terms in the context of *Decknamen*. By annotating occurrences of all *Decknamen* in the ontology, a network of *Decknamen* is created. Finally, the terms in the alchemical texts are linked to the ontology and the concordance, which, in turn, contextualise alchemical language.

To take the alchemical term "Mercurius" as an example of this process, the *word* "Mercurium" links to the *concept* "Mercurius," which can have chemical, mythological, and/or historical contexts. The "Keyword in Context" of *Decknamen* will suggest which of those possible contexts is most prevalent in the surrounding paragraphs, and thus a probable disambiguation can be made. Figure 1 shows an occurrence of "Mercurius" of which possible contexts could be "planets," "metals," and "mythological." Since the *Decknamen* surrounding this term could also be contextualised as planets or metals, it is likely that in this case "Mercurius" is meant as a planet or chemical agent and not as a mythological figure.

42 *Continued*
contextualise them. It is a hermeneutical tool for the study of texts to enrich traditional approaches to textual analysis rather than a method to replace them. See Franco Moretti, *Distant Reading* (London: Verso, 2013); Matthew L. Jockers, *Macroanalysis: Digital Methods and Literary History* (Chicago: Chicago University Press, 2013); Stéfan Sinclair and Geoffrey Rockwell, *Hermeneutica: Computer-Assisted Interpretation in the Humanities* (Cambridge, MA: MIT Press, 2016).

43 See, for example Frietsch, "Alchemie Thesaurus."

44 Here, KOS means a digital resource (knowledge organisation system) coded in a combination of RDFS, SKOS, and XML that can be used as a hermeneutic and analytical tool. It does not aim to offer keywords to tag library catalogues, nor does it provide universal definitions for lemmata. It is not meant as an introductory source, but a resource for the analysis of a textual corpus. The purpose of this KOS is not to collect and condense all current knowledge on a topic in short entries, but to function as a digital ontology to describe classes and relationships. It is thus fundamentally different from a dictionary (which describes words in terms of linguistics) and an encyclopaedia (which collects encyclopaedic knowledge about a subject or its respective headwords).

```
<example ref="Maier.Arcana.191">
  tum, aut per <label>Lunam</deckname>, <deckname>argentum</deckname>,
  per <termInQuestion>Mercurium</termInQuestion>
  <deckname>hydrargyrum</deckname>, per <deckname>Saturnum</deckname>
  <deckname>plumbum</deckname>, per <deckname>Iovem</deckname>,  <deckname>stannum</deckname>,
  per <deckname>Martem</deckname> <deckname>ferrum</deckname>, communia intellexis
</example>
```

FIGURE 1 An example of a "Context of Decknamen" formatted in pseudo-code XML. XML, the eXtensible Markup Language, encodes data in a human and machine-readable format, providing, for example, standardised encoding using the Text Encoding Initiative (TEI) XML standard. To improve readability, XML and RDF examples in this article are reduced to simplified pseudo-code versions containing only relevant elements for demonstration purposes.

```
:PhilosophersStone :hasColor :red .
:red :hasChemicalProperty :tints .
:tints :givesPhysicalProperty :citrinitas .
:Gold :hasColor :citrinitas .
```

FIGURE 2 Example of parts of a KOS encoded as RDF-Schema (RDFS) triples.

A step further would be to model this together with other knowledge about alchemical analogies (Figure 2). A quantitative analysis based on a corpus, where the *Decknamen* have been annotated, allows us to trace their relationships to other terms and create probabilities illustrating how likely it is for a word to appear in a specific context based on the other terms clustered around it and their possible concepts. However, digital models, as Willard McCarty points out, are "temporary states in a process of coming to know" in which computers are not "knowledge jukeboxes" but "representation machines" and hermeneutic tools.[45] The probabilities they offer are not scholarly interpretations but data. Extracting contexts from words using a concordance taken from the corpus itself allows one book to explain another, which is the mode of distribution of alchemical knowledge.[46]

Michael Maier's opus is a good candidate for machine reasoning methods, as he employs alchemical secretive language devices and explains himself with a good degree of redundancy over the whole of his corpus.[47] This allows for an algorithm to compare the many uses and contexts of specific terms such as "Mercurius" within the vast corpus of Maier. Widely available word-count-based methods and

45 Willard McCarty, "Modeling: A Study in Words and Meanings," in *A Companion to Digital Humanities*, eds. Susan Schreibmann, Ray Siemens, and John Unsworth (Oxford: Wiley Blackwell, 2004), 254–72, on 255.

46 Another function of the method presented could be to detect instances of the alchemical cryptographical device referred to as "dispersion of knowledge."

47 This method will be published as a digital edition of Michael Maier's corpus in GAMS infrastructure (gams.uni-graz.at) so that readers can try it out for themselves and, most importantly, use it as a catalyst to advance and speed up otherwise time-consuming research on Maier's work. After that, it would be desirable to expand the knowledge base coded in the digital ontology and apply it to other alchemical sources.

other forms of quantitative research only provide *data*, not interpretation.[48] The example of "Mercurius" occurs very frequently and covers an enormous range of possible meanings. A quantitative text analysis approach would give all mentions of "Mercurius" the same weight and meaning, when in reality this is not the case. Thus, different possible meanings must be disambiguated using *close reading*.

Visualising dragon's blood

Another case study from the Maier corpus demonstrates the possibilities of such an approach. In his *Viatorium* (1618), an allegory of "dragon's blood" concludes the chapter on Saturnus.[49] Maier employs the alchemical technique of parathesis by enumerating a multitude of mythological allusions.[50] A KOS that connects all terms related to a snake or dragon (from the Latin *draco*), can demonstrate how different names for the same thing are being enumerated, possibly to confuse an incompetent reader. The chapter includes an allegory called *Sanguinis Draconis Descriptio*, meant as an etiological narrative to explain the origin of the alchemical dragon's blood. The allegorical function and situation towards the end of the chapter are to support Maier's theory presented earlier in the chapter. In this context, Maier is also using the allegory as a parathesis to enumerate a host of *Decknamen* in order to confuse the uninitiated reader. The method proposed in this paper can be illustrated by investigating the grouping of alchemical terms around "*Saturno*" in Maier's description of dragon's blood. For digital analysis to be possible, the computer needs to be instructed on which words should be considered. This is done through digital annotation in XML (Figure 3).

Next, the semantic context must be determined. The KOS can be given multiple possible contexts for each term that can then be encoded following the RDFS standard to show class relationships. For example, the concept ":draco," defined as having ":contextAllegorical," appears twice in the short excerpt. There is also one instance of ":elephas" with a shared context, yielding a total of three hits for ":contextAllegorical" in this particular passage. All animals were tagged as possible allegories to expedite the analysis. Since the digital algorithm only supplies data that can be used for interpretation but cannot perform an interpretation itself, each label doesn't need to be one-hundred per cent accurate. Instead, the quality of the model should be evaluated based on the usefulness of the predictions it supplies.

Theoretically, there are infinite possible contexts for a term. However, to create a working model, the number of alchemical terms surrounding "*Saturno*" was limited to fifteen present concordances. As mentioned previously, "*Saturno*" yields one count each for ":contextMythological," ":contextChemical," ":contextMetals," and ": contextPlanets." The mythological terms Maier enumerates give us a total of seven ":contextMythological." Statistically, the dominant context of our model is

[48] Franco Moretti, *Graphs, Maps, Trees: Abstract Models for a Literary History* (London: Verso, 2005), 9.

[49] Michael Maier, *Viatorium* (Oppenheim, 1618), 54–6.

[50] William Newman defines parathesis as: "the heaping-up of synonyms for a given process, substance, or apparatus, again with the intention of bewildering the reader." Newman, "'Decknamen or Pseudochemical Language'?" 180.

```
<example n="1" ref="Maier.Viatorium.55">
    … qui post <term>DRACONIS</term> (licet potior pars <term>elephantis</term> sit)
    dicitur, ad <term>medicinam</term> aliasque necessitates hominum non inutilis.
    <term>Draconem</term> hunc, cujus <term>sanguis</term> ad <term>medicinam
    philosophicam <term> assumitur,
    oportet quem probè agnoscere, ne <pb n="56"/>
    <term>terreum</term> quid, seu ex <term>terrae</term> fundo depromptum,
    aut aliud haud conveniens in ejus locum substituat.
    Est autem ex eo genere, quod <deckname>Saturno</deckname>,
    <term>Apollini</term>, <term>Aesculapio</term>
    et <term>Mercurio</term> asscribitur, quod <term>Cadmi</term> socios interfecit,
    et ab eo interfectum, quod <tem>vellus aureum</tem> observavit, quod <term>pomis aureis</term>
    <term>Hesperidum hortis</term> invigilavit,
</example>1
```

FIGURE 3 Example of a digital annotation in XML. From the passage Maier, *Viatorium*, 55–56: "Its blood is then called [dragon's / snake's] blood (even though it is rather part of the elephant), quite useful to medicine and other necessities of humankind. This snake whose blood is taken as philosophical medicine needs to be properly understood, that is, not as something earthly taken from the ground, so that it is not wrongly substituted. It is of the kind which is attributed to Saturn, to Apollo, to Aesculapius and Mercurius. It is that which killed Cadmos' comrades and was killed by him, which guarded the Golden Fleece and the apples from the Garden of the Hesperides […]." (Translated into English by the author).

mythological, and all concepts occurring after "*Saturno*" can be understood in a mythological context (Tables 1 and 2). Additionally, the knowledge base encoded in the digital ontology shows a close connection to the concept of ":draco." The statistics point the computer to the fact that there seems to be a parathesis here, indicated by the sudden and localised shift in the context in combination with the mythological terms all having a singular concept of ":draco" in common, which is the concept Maier discussed and allegorised at length in the preceding section.

By situating a digital ontology as central to the methodology, it is now possible to automatically recognise the allegory of "*dracones*" even without the term being explicitly mentioned.[51] Readers familiar with alchemical interpretation might have noticed this connection through close reading, but making only one example clearer is not the purpose of such an algorithm. The algorithm can detect patterns over a larger corpus of texts and direct readers to parts of a text that are of particular interest without the labour of close reading. Additionally, machine reasoning can highlight potential instances of intertextuality and dispersion of knowledge by drawing connections between localised text passages that are very similar to each other in content. For a corpus the size of Maier's, this would take quite a long time to do with only close reading. A digital tool such as the one discussed here allows scholars to focus their time and energy on aspects of research that require human intervention, like interpreting alchemical language.

51 Mythological beasts like Python slayed by Apollo or Ladon guarding the Garden of the Hesperides are usually referred to by their names rather than by the word *draco*.

TABLE 1

TABLE VISUALISING WHICH WORDS FROM THE TEXT (ROWS) GET ASSIGNED WHICH CONTEXTS (COLUMNS) AND THE RESULT. THIS TABLE ILLUSTRATES HOW ONE TERM CAN BE ASSIGNED MULTIPLE CONTEXTS/CATEGORIES, THE EXTENT OF WHICH CAN NOT BE FULLY RENDERED IN A TWO-DIMENSIONAL VIEW

	:Mytholog	:Allegor.	:Chemical	:Elements	:Iatrochyn	:Planets	:Metals	:Alch.Thec	:Alch.Trad	:Historical
DRACONIS		x								
elephantis		x								
medicinam					x					
Draconem		x								
sanguis			x							
medicinam philosophicam					x					
terreum				x						
terrae				x						
Saturno	x		x			x	x			
Apollini	x									
Aesculapio	x									
Mercurio	x		x			x	x	x	x	x
Cadmi	x									
vellus aureum	x									
pomis aureis	x									
Hesperidum hortis	x									
SUM	8	3	3	2	2	2	2	1	1	1

TABLE 2

TABLE SHOWING THE TEXT AS PSEUDO-CODE MARKUP. THE ANNOTATED TERMS ARE SHOWN IN COLOUR ACCORDING TO THE MAIN CATEGORY ASSIGNED TO THEM AND VISUALISE HOW THE CONTEXT SWITCHES TO PREDOMINATELY MYTHOLOGICAL TOWARDS THE SECOND HALF OF THE TEXT SECTION

<example ref="Maier.Viatorium.55»

… qui post [allegorical] DRACONIS (licet potior pars [allegorical] elephantis sit) dicitur, ad [iatrochymistry] medicinam aliasque

necessitates hominum non inutilis. [allegorical] Draconem hunc, cujus [chemical] sanguis ad

[iatrochymistry] medicinam philosophicam assumitur, oportet quem probè agnoscere, ne <pb n="56/> [elements] terreum

quid, seu ex [elements] terrae fundo depromptum, aut aliud haud conveniens in ejus locum substituat. Est

autem ex eo genere, quod <deckname>Saturno</deckname>, [mythological] Apollini, [mythological] Aesculapio et

[mythological] Mercurio asscribitur, quod [mythological] Cadmi socios interfecit, et ab eo interfectum, quod [mythological] vellus aureum

observavit, quod [mythological] pomis aureis [mythological] Hesperidum hortis invigilavit, </example>

Conclusion

An increasing number of alchemical and chymical texts have become available online as digital facsimiles over the past years.[52] The recent *Furnace and Fugue*

[52] As of 29 April 2020, a VD-17 search ("Gattungsbegriffe (Alte Drucke) alchem*") yields 1339 hits, of which 953 are digital facsimiles of print from the German-speaking areas in the seventeenth century. Cf. DFG, "VD17 – Das

project offers a beautiful digital edition of Maier's *Atalanta fugiens*; however, the project does not employ quantitative text analysis.[53] Similarly, in his 2011 prognosis on the future of alchemy research, Marcos Martinón-Torres mentions the digital sphere, but only as a means of making textual sources more widely accessible to provide digital platforms, and not for its usefulness in the analysis of alchemical texts.[54] However, when digital texts become available, digital analysis (beyond simple querying) becomes possible too.[55] Scholars have yet to investigate the possibilities of digital methods as tools for the analysis of the ever-growing amount of digitised alchemical texts. The digital humanities offer a wide range of methods, many of which could be useful in addressing relevant questions in the historiography of alchemy. However, not all digital methods can be applied to the analysis of the phenomenon of alchemical language specifically, as discussed in this paper.[56] For texts to be used in digital text processing, digital facsimiles must first be transcribed as digital texts. This can be accomplished manually or with specialised software such as the *Transkribus* Handwritten Character Recognition Software, which works remarkably well on early modern print.[57]

Studies put forth by Principe and Newman have proposed that we use knowledge of chemical processes to decode and analyse alchemical texts containing *Decknamen*. The recreation of alchemical experiments in a modern laboratory serves as proof of coherence for the textual interpretation.[58] However, similar ends can be achieved using computerised machine reasoning. Alchemical language relies heavily on analogies, and this type of analogical communication through chains of associations can easily be modelled digitally. Existing text mining methods only allow for the compilation of data but cannot answer specific queries

52 *Continued*
Verzeichnis der im deutschen Sprachraum erschienenen Drucke des 17. Jahrhunderts," 2017, http://www.vd17.de/ (accessed 3 February 2021).

53 Nummedal and Bilak, *Furnace and Fugue*.

54 Martinón-Torres, "Some Recent Developments," 233.

55 Jockers criticises this "disciplinary habit of thinking small: the traditionally minded scholar recognizes value in digital texts because they are individually searchable, but this same scholar, as a result of a traditional training, often fails to recognise the potentials for analysis that an electronic processing of text enables." Jockers, *Macroanalysis*, 17.

56 Stylometric analysis, i.e. comparing so-called "author signals," can be understood as the search for a sort of statistical fingerprint by which unique authors can be distinguished. Stylometry is a quantitative, statistical method applied to humanities questions developed before the advent of the computer, but can now be conducted computationally. For example, see Lang, "Assessing Michael Maier's Contributions to Francis Anthony's Apologia (1616) Using Stylometry," in *Proceedings of the Conference on Computational Humanities Research 2021*, ed. Maud Ehrmann et al. (Amsterdam: CEUR Workshop Proceedings, 2021), 346–58, http://ceur-ws.org/Vol-2989/short_paper44.pdf (accessed 25 January 2022).

57 For a machine learning based software to be able to automatically perform a given task, it needs to "learn" from a dataset. One result from this is a model of how the program "has learned" to deliver the expected outputs. A freely available model suitable for the automated transcription of early modern Neo-Latin print is the Noscemus Genereal Model version 4 (Noscemus GM v4). Günther Mühlberger, "Transkribus," https://transkribus.eu/Transkribus/ (accessed 3 February 2021). On how to use Transkribus for alchemical sources see Michael Fröstl, Stefan Zathammer and Sarah Lang, "Zur Transkription von Alchemica mithilfe der Transkribus-Software," in *Alchemistische Labore. Praktiken, Texte und materielle Hinterlassenschaften / Alchemical Laboratories: Practices, Texts, Material Relics*, ed. Sarah Lang, Michael Fröstl and Patrick Fiska (Graz: Grazer Universitätsverlag, 2022).

58 Lawrence M. Principe, "Robert Boyle's Alchemical Secrecy: Codes, Ciphers and Concealments," *Ambix* 39 (1992): 63–75; Newman, "'*Decknamen* or Pseudochemical Language'?"

investigating the phenomenon of alchemical language and its *Decknamen*.[59] Simply querying a single *Deckname* through full-text search or quantitative textual analysis only yields helpful results if the term in question is not too frequent. When in reality, most queries (such as "Mercurius," which is mentioned over 1000 times in Maier's opus alone) involve complex terms that require a method for automated disambiguation and contextualisation.

This paper has proposed a digital algorithm for the application of the methodology developed in the "New Historiography of Alchemy" for the decoding of alchemical *Decknamen*. Further research is needed to address the issue of integrating historical and modern alchemical dictionaries in the form of Linked Open Data. The time and effort needed to analyse *Decknamen* through large-scale distant reading can be reduced by using digital tools that can scan vast texts in seconds to reveal clusters of meanings and link terms. In this way, the method proposed in this essay functions as a catalyst for research – rather than presenting an approach that is new and untested, it is based on tried and tested methodology from scholars in the field and offers a digital means of accelerating the analytical process. How can scholars apply digital methodologies that transcend simply digitising material resources or virtually representing the traditional print paradigm despite being born-digital resources to create a computational approach to the analysis of alchemical language in the future? The application of machine learning algorithms to the decoding of alchemical language could yield promising results.

Acknowledgements

I thank Megan Piorko for the organisation of the SHAC Postgraduate Workshop in Amsterdam 2019 and her support in the publication process and the anonymous reviewers whose comments helped improve this paper. Additionally, I thank the Gotha Netzwerk Alchemie for their valuable feedback, especially Rainer Werthmann, for his indispensable help with chemical interpretations.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research is part of my Ph.D. project, which was generously funded by the University of Graz, the Rolf und Ursula Schneider-Stiftung zur Förderung der Geschichtswissenschaften, which funded a research stay at Herzog August

[59] The whole opus *Maierianum* was machine transcribed using the Transkribus Optical Character Recognition. Even though this corpus was neither double-key corrected or cleaned up, with a rate of 1–2 errors per page it is of sufficient quality to obtain consistent results when analysing it in combination with the web-based reading and analysis environment for digital texts, Voyant Tools (https://voyant-tools.org/).

Bibliothek Wolfenbüttel and Leibniz Institut für Europäische Geschichte Mainz for a Digital Humanities fellowship conducted as part of this project.

Note on contributor

Sarah Lang studied Classics and History in Graz and Montpellier. She is Digital Humanities Postdoctoral Fellow at Centre for Information Modelling (ZIM) Graz. In her PhD thesis in Digital Humanities, she developed a machine reasoning algorithm and semantic web-based analysis tool for alchemical *Decknamen* on the Neo-Latin corpus of Michael Maier. She has been a fellow at German Historical Institute Paris, Herzog August Bibliothek Wolfenbüttel, Leibniz Institut für europäische Geschichte Mainz, Ludwig Boltzmann Institute for Neo-Latin Studies Innsbruck, and the Science History Institute in Philadelphia. Zentrum für Informationsmodellierung (Centre for Information Modelling), University of Graz, Elisabethstraße 59/III, 8010 Graz, Steiermark, Austria. Email: sarah.lang@uni-graz.at.

ORCID

Sarah Lang http://orcid.org/0000-0002-4618-9481